THE KUMMER PAIRING AND WEAK MORDELL–WEIL, WITH SOLUTIONS

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The weak Mordell–Weil theorem says that $E(K)/mE(K)$ is finite for an elliptic curve E over a number field K. The proof in [Sil09, §VIII.1] has three steps. The Kummer pairing turns the finiteness of $E(K)/mE(K)$ into the finiteness of the field L generated by the m-th divisors of the rational points. This field is abelian of exponent m and unramified outside a finite set of places S. And Kummer theory shows that such fields are finite. This worksheet runs the argument for $m = 2$ and $K = \mathbb{Q}$, on curves whose points of order two are rational. Then the class group and the S-unit theorem disappear, and the injectivity of reduction on the 2-torsion becomes elementary. Problem 1.6 sets up the pairing. Problem 2.7, the heart of the worksheet, proves that L is unramified outside S and deduces a bound for $E(\mathbb{Q})/2E(\mathbb{Q})$. Problem 3.1 tests the bound on $y^2 = x^3 - x$. Each problem is preceded by the definitions and results it uses.

Standing notation.

- An elliptic curve E over $\mathbb{Q}$, given by
$$y^2 = (x - e_1)(x - e_2)(x - e_3)$$
with distinct $e_1, e_2, e_3 \in \mathbb{Z}$, and its points of order two $T_i := (e_i, 0)$. So $E[2](\overline{\mathbb{Q}}) = \{O, T_1, T_2, T_3\} \subseteq E(\mathbb{Q})$.
- The finite set of primes $S := \{2\} \cup \{p : p \mid (e_1 - e_2)(e_1 - e_3)(e_2 - e_3)\}$, and $s := \#S$.
- The absolute Galois group $\mathrm{Gal}_{\mathbb{Q}} := \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, with elements σ, τ, γ. For an extension F of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$, $\mathrm{Gal}_F := \mathrm{Gal}(\overline{\mathbb{Q}}/F)$.
- For $Q \in E(\overline{\mathbb{Q}})$, the field $\mathbb{Q}(Q)$ generated by the coordinates of Q, and the point $\sigma(Q)$, obtained by applying $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$ to the coordinates.
- The compositum L of the fields $\mathbb{Q}(Q)$ for $Q \in E(\overline{\mathbb{Q}})$ with $2Q \in E(\mathbb{Q})$. Silverman writes $L = \mathbb{Q}([2]^{-1}E(\mathbb{Q}))$.
- The quotient $B/2 := B/2B$ of an abelian group B, and the group $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ of classes of rational numbers modulo squares.

How to use this. Allow about 20 minutes for Problem 1.6, 35 for Problem 2.7 and 25 for Problem 3.1. Problem 2.7 is the one not to skip. If cohomology has not come up yet in lecture, leave Problem 1.6(4) for later. The `Magma` block runs in seconds in the free online calculator. Everything else is by hand.

1. The Kummer pairing

This section is [Sil09, Proposition VIII.1.2], for $m = 2$ and $K = \mathbb{Q}$, and its reinterpretation in [Sil09, §VIII.2], with cohomology as in [Sil09, §§B.1–B.2].

Definition 1.1 (The Kummer pairing [Sil09, §VIII.1])**.** The Kummer pairing is the map
$$\kappa\colon E(\mathbb{Q}) \times \mathrm{Gal}_{\mathbb{Q}} \to E[2](\overline{\mathbb{Q}})$$
defined as follows. Let $P \in E(\mathbb{Q})$ and $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$. Choose $Q \in E(\overline{\mathbb{Q}})$ with $2Q = P$, and set $\kappa(P, \sigma) := \sigma(Q) - Q$. That this lands in $E[2](\overline{\mathbb{Q}})$, and does not depend on Q, is Problem 1.6(1).

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Proposition 1.2 (Infinite Galois theory [Mil22, Theorem 7.13]). *For a closed subgroup H of $\mathrm{Gal}_{\mathbb{Q}}$, let $\overline{\mathbb{Q}}^H$ be its fixed field. The maps $H \mapsto \overline{\mathbb{Q}}^H$ and $F \mapsto \mathrm{Gal}_F$ are inverse bijections between the closed subgroups of $\mathrm{Gal}_{\mathbb{Q}}$ and the extensions of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$. The subgroup H is normal if and only if $\overline{\mathbb{Q}}^H$ is Galois over $\mathbb{Q}$, and then $\mathrm{Gal}(\overline{\mathbb{Q}}^H/\mathbb{Q}) \cong \mathrm{Gal}_{\mathbb{Q}}/H$ by restriction. In particular, $\overline{\mathbb{Q}}^{\mathrm{Gal}_{\mathbb{Q}}} = \mathbb{Q}$.*

Definition 1.3 (Cohomology in degree 1 [Sil09, §§B.1–B.2]). Let A be a discrete $\mathrm{Gal}_{\mathbb{Q}}$-module. A 1-cocycle is a continuous map $\xi\colon \mathrm{Gal}_{\mathbb{Q}} \to A$, $\sigma \mapsto \xi_\sigma$, with $\xi_{\sigma\tau} = \xi_\sigma + \sigma\xi_\tau$ for all σ, τ. For $a \in A$, the map $\sigma \mapsto \sigma a - a$ is a 1-cocycle, and the 1-cocycles of this form are the 1-coboundaries. Then $H^1(\mathrm{Gal}_{\mathbb{Q}}, A)$ is the group of 1-cocycles modulo the 1-coboundaries. For an elliptic curve or its 2-torsion, we abuse notation and write $H^1(\mathbb{Q}, E[2]) \coloneqq H^1(\mathrm{Gal}_{\mathbb{Q}}, E[2](\overline{\mathbb{Q}}))$.

Proposition 1.4 (The long exact sequence [Sil09, Propositions B.1.2 and B.2.3]). *Let $0 \to A \to B \xrightarrow{\psi} C \to 0$ be an exact sequence of discrete $\mathrm{Gal}_{\mathbb{Q}}$-modules. For $c \in C^{\mathrm{Gal}_{\mathbb{Q}}}$, choose $b \in B$ with $\psi(b) = c$, and let $\delta(c) \in H^1(\mathrm{Gal}_{\mathbb{Q}}, A)$ be the class of $\sigma \mapsto \sigma b - b$. This does not depend on b, and the sequence*

$$0 \to A^{\mathrm{Gal}_{\mathbb{Q}}} \to B^{\mathrm{Gal}_{\mathbb{Q}}} \to C^{\mathrm{Gal}_{\mathbb{Q}}} \xrightarrow{\delta} H^1(\mathrm{Gal}_{\mathbb{Q}}, A) \to H^1(\mathrm{Gal}_{\mathbb{Q}}, B)$$

is exact.

Proposition 1.5 (Trivial action [Sil09, Remark B.2.2]). *If $\mathrm{Gal}_{\mathbb{Q}}$ acts trivially on A, then*

$$H^1(\mathrm{Gal}_{\mathbb{Q}}, A) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_{\mathbb{Q}}, A),$$

the group of continuous homomorphisms.

Problem 1.6 (The Kummer pairing).

(1) Show that $\kappa(P, \sigma) \in E[2](\overline{\mathbb{Q}})$, and that it does not depend on the choice of Q.

Solution. The multiplication map $[2]$ is given by rational functions with coefficients in $\mathbb{Q}$, so it commutes with σ. Hence

$$2\big(\sigma(Q) - Q\big) = \sigma(2Q) - 2Q = \sigma(P) - P = O,$$

as P is a rational point. Any other choice is $Q + T$ with $T \in E[2](\overline{\mathbb{Q}})$, and $\sigma(T) = T$ because $T \in E(\mathbb{Q})$. So $\sigma(Q+T) - (Q+T) = \sigma(Q) - Q$.

(2) Let $P, P' \in E(\mathbb{Q})$ and $\sigma, \tau \in \mathrm{Gal}_{\mathbb{Q}}$. Show that $\kappa(P + P', \sigma) = \kappa(P, \sigma) + \kappa(P', \sigma)$ and

$$\kappa(P, \sigma\tau) = \kappa(P, \sigma) + \sigma\big(\kappa(P, \tau)\big).$$

Deduce that κ is bilinear. Where does the hypothesis $E[2](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q})$ enter?

Solution. **In the independence of Q, and to remove σ from $\sigma(\kappa(P, \tau))$.** The addition law is given by rational functions over $\mathbb{Q}$, so σ acts on $E(\overline{\mathbb{Q}})$ by group automorphisms. If $2Q = P$ and $2Q' = P'$, then $2(Q+Q') = P+P'$, and $\sigma(Q+Q') - (Q+Q') = (\sigma(Q) - Q) + (\sigma(Q') - Q')$. For the second identity,

$$(\sigma\tau)(Q) - Q = \big(\sigma(Q) - Q\big) + \sigma\big(\tau(Q) - Q\big).$$

This is the cocycle identity of Definition 1.3, and it holds for every E. Here $\kappa(P, \tau) \in E[2](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q})$ is fixed by σ, so $\kappa(P, \sigma\tau) = \kappa(P, \sigma) + \kappa(P, \tau)$, and $\kappa(P, \cdot)$ is a homomorphism. The hypothesis is used twice: for the independence of Q in (1), and here.

(3) Show that $\kappa(P,\cdot) = O$ if and only if $P \in 2E(\mathbb{Q})$, and that $\kappa(\cdot,\sigma) = O$ if and only if $\sigma \in \mathrm{Gal}_L$. Deduce that $L/\mathbb{Q}$ is Galois and that κ induces a perfect pairing

$$E(\mathbb{Q})/2 \times \mathrm{Gal}(L/\mathbb{Q}) \to E[2](\overline{\mathbb{Q}}),$$

that is, both $E(\mathbb{Q})/2 \to \mathrm{Hom}(\mathrm{Gal}(L/\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$ and $\mathrm{Gal}(L/\mathbb{Q}) \to \mathrm{Hom}(E(\mathbb{Q})/2, E[2](\overline{\mathbb{Q}}))$ are injective. Conclude that $E(\mathbb{Q})/2$ is finite if and only if $L/\mathbb{Q}$ is finite.

Solution. If $P = 2Q$ with $Q \in E(\mathbb{Q})$, then $\sigma(Q) = Q$ for all σ. Conversely, if $\sigma(Q) = Q$ for all σ, then the coordinates of Q lie in $\overline{\mathbb{Q}}^{\mathrm{Gal}_\mathbb{Q}} = \mathbb{Q}$ (Proposition 1.2), so $P = 2Q \in 2E(\mathbb{Q})$. Next, if $\sigma \in \mathrm{Gal}_L$ and $2Q \in E(\mathbb{Q})$, then $Q \in E(L)$, so $\sigma(Q) = Q$. Conversely, if $\kappa(\cdot,\sigma) = O$, then $\sigma(Q) = Q$ for every Q with $2Q \in E(\mathbb{Q})$, so σ fixes each $\mathbb{Q}(Q)$, hence their compositum L, and $\sigma \in \mathrm{Gal}_L$ by Proposition 1.2.
By (2), $\sigma \mapsto \kappa(\cdot,\sigma)$ is a homomorphism $\mathrm{Gal}_\mathbb{Q} \to \mathrm{Hom}(E(\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$ with kernel Gal_L, so Gal_L is normal and $L/\mathbb{Q}$ is Galois (Proposition 1.2). The pairing factors through $E(\mathbb{Q})/2$ on the left and through $\mathrm{Gal}_\mathbb{Q} / \mathrm{Gal}_L \cong \mathrm{Gal}(L/\mathbb{Q})$ on the right, and the two kernels computed above say that both induced maps are injective. If $E(\mathbb{Q})/2$ is finite, then $\mathrm{Gal}(L/\mathbb{Q})$ injects into the finite group $\mathrm{Hom}(E(\mathbb{Q})/2, E[2](\overline{\mathbb{Q}}))$. If $L/\mathbb{Q}$ is finite, then $E(\mathbb{Q})/2$ injects into the finite group $\mathrm{Hom}(\mathrm{Gal}(L/\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$.

(4) Show that $0 \to E[2](\overline{\mathbb{Q}}) \to E(\overline{\mathbb{Q}}) \xrightarrow{2} E(\overline{\mathbb{Q}}) \to 0$ is an exact sequence of discrete $\mathrm{Gal}_\mathbb{Q}$-modules, and that its connecting map $\delta\colon E(\mathbb{Q}) \to H^1(\mathbb{Q}, E[2])$ sends P to the class of $\kappa(P,\cdot)$. Show that $H^1(\mathbb{Q}, E[2]) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_\mathbb{Q}, E[2](\overline{\mathbb{Q}}))$, and recover from Proposition 1.4 the injectivity of $E(\mathbb{Q})/2 \to \mathrm{Hom}(\mathrm{Gal}_\mathbb{Q}, E[2](\overline{\mathbb{Q}}))$.

Solution. The map $[2]$ is a nonconstant morphism of curves, so it is surjective on $\overline{\mathbb{Q}}$-points [Sil09, Proposition II.2.3], and its kernel is $E[2](\overline{\mathbb{Q}})$ by definition. Every point of $E(\overline{\mathbb{Q}})$ has coordinates in a finite extension F of $\mathbb{Q}$, so its stabilizer contains the open subgroup Gal_F, and the module is discrete. By Proposition 1.4, $\delta(P)$ is the class of $\sigma \mapsto \sigma(Q) - Q$ with $2Q = P$, which is $\kappa(P,\cdot)$. As $\mathrm{Gal}_\mathbb{Q}$ acts trivially on $E[2](\overline{\mathbb{Q}})$, Proposition 1.5 identifies $H^1(\mathbb{Q}, E[2])$ with $\mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_\mathbb{Q}, E[2](\overline{\mathbb{Q}}))$, a group in which every coboundary is 0. The exact sequence of Proposition 1.4 reads

$$E(\mathbb{Q}) \xrightarrow{2} E(\mathbb{Q}) \xrightarrow{\delta} H^1(\mathbb{Q}, E[2]),$$

so $\ker\delta = 2E(\mathbb{Q})$, which is the first half of (3).

Remark. Without the hypothesis $E[2](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q})$, the map δ is still defined and injective on $E(\mathbb{Q})/2$, but $H^1(\mathbb{Q}, E[2])$ is no longer a group of homomorphisms. Silverman reduces to the case of rational torsion by [Sil09, Lemma VIII.1.1.1], or by inflation–restriction [Sil09, §VIII.2]. The next group in the long exact sequence, $H^1(\mathbb{Q}, E)$, measures the failure of δ to be surjective, and leads to the Selmer and Shafarevich–Tate groups [Sil09, §X.4].

2. Unramified outside S

This section is [Sil09, Proposition VIII.1.5], for $m = 2$ and $K = \mathbb{Q}$. Its argument is that of [Sil09, Proposition VII.4.1(a)], applied to the halves of rational points instead of torsion points. The last item adds Kummer theory, as in [Sil09, proof of Proposition VIII.1.6]. In this section p is a prime, and $\overline{\mathbb{Q}}$ is embedded in an algebraic closure $\overline{\mathbb{Q}}_p$ of $\mathbb{Q}_p$, so that restriction is an injection $\mathrm{Gal}_{\mathbb{Q}_p} \coloneqq \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \hookrightarrow \mathrm{Gal}_\mathbb{Q}$. The residue field of the valuation ring of $\overline{\mathbb{Q}}_p$ is an algebraic closure $\overline{\mathbb{F}}_p$ of $\mathbb{F}_p$, and a bar denotes reduction modulo its maximal ideal.

Definition 2.1 (Inertia [Sil09, §VII.4])**.** An element $\sigma \in \mathrm{Gal}_{\mathbb{Q}_p}$ preserves the valuation ring of $\overline{\mathbb{Q}}_p$, so it induces an automorphism $\bar\sigma$ of $\overline{\mathbb{F}}_p$. The inertia group $I_p \subseteq \mathrm{Gal}_{\mathbb{Q}_p}$ is the subgroup of the σ with

$\bar{\sigma} = 1$. If ${\mathbb{Q}_p}^{\mathrm{unr}} \subseteq \overline{\mathbb{Q}}_p$ is the maximal unramified extension of $\mathbb{Q}_p$, then $I_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/{\mathbb{Q}_p}^{\mathrm{unr}})$. A Galois extension F of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$ is unramified at p if $I_p \subseteq \mathrm{Gal}_F$, that is, if I_p fixes F.

Proposition 2.2 (Good reduction [Sil09, §III.1 and Remark VIII.1.3])**.** *The Weierstrass equation of E has discriminant*

$$\Delta = 16(e_1 - e_2)^2(e_1 - e_3)^2(e_2 - e_3)^2.$$

If $p \nmid \Delta$, the equation is minimal at p and its reduction E_p is an elliptic curve over $\mathbb{F}_p$.

Proposition 2.3 (Reduction [Sil09, Proposition VII.2.1])**.** *Suppose that $p \nmid \Delta$. Every $Q \in E(\overline{\mathbb{Q}}_p)$ has projective coordinates $[X : Y : Z]$ in the valuation ring of $\overline{\mathbb{Q}}_p$, not all in its maximal ideal, and $\bar{Q} := [\bar{X} : \bar{Y} : \bar{Z}] \in E_p(\overline{\mathbb{F}}_p)$. The reduction map $E(\overline{\mathbb{Q}}_p) \to E_p(\overline{\mathbb{F}}_p)$, $Q \mapsto \bar{Q}$, is a homomorphism, and $\overline{\sigma(Q)} = \bar{\sigma}(\bar{Q})$ for $\sigma \in \mathrm{Gal}_{\mathbb{Q}_p}$.*

Proposition 2.4 (Kummer theory [Sil09, Exercise 8.4])**.** *Let F be a Galois extension of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$ with $\mathrm{Gal}(F/\mathbb{Q})$ abelian of exponent dividing 2. Let $D \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ be the subgroup of the classes of the $d \in \mathbb{Q}^\times$ that are squares in F. Then $F = \mathbb{Q}(\sqrt{d} : d \in D)$.*

Proposition 2.5 (Unramified quadratic extensions [Sil09, proof of Proposition VIII.1.6])**.** *Let p be an odd prime and $d \in \mathbb{Q}^\times$. Then $\mathbb{Q}(\sqrt{d})$ is unramified at p if and only if $v_p(d)$ is even.*

Definition 2.6 (The group $\mathbb{Q}(S, 2)$ [Sil09, Theorem X.1.1(c)])**.** The subgroup $\mathbb{Q}(S, 2) \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ consists of the classes of the $d \in \mathbb{Q}^\times$ with $v_p(d)$ even for every prime $p \notin S$. It is generated by the classes of -1 and of the primes in S, so it has order 2^{s+1}.

Problem 2.7 ($L/\mathbb{Q}$ is unramified outside S)**.**

(1) Show that $\mathrm{Gal}(L/\mathbb{Q})$ is abelian of exponent dividing 2. Let $D \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ be the subgroup of the classes of the $d \in \mathbb{Q}^\times$ that are squares in L. Deduce that $L = \mathbb{Q}(\sqrt{d} : d \in D)$.

Solution. By Problem 1.6(2) and (3), $\sigma \mapsto \kappa(\cdot, \sigma)$ is an injective homomorphism $\mathrm{Gal}(L/\mathbb{Q}) \hookrightarrow \mathrm{Hom}(E(\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$. The target is abelian and killed by 2, so the same holds for $\mathrm{Gal}(L/\mathbb{Q})$. As $L/\mathbb{Q}$ is Galois, Proposition 2.4 applies to $F = L$.

(2) Let $p \notin S$. Show that E has good reduction at p, and that O, T_1, T_2, T_3 reduce to four distinct points of $E_p(\mathbb{F}_p)$. Deduce that reduction is injective on $E[2](\overline{\mathbb{Q}})$.

Solution. The prime divisors of Δ are 2 and the primes dividing some $e_i - e_j$, that is, the primes in S. So $p \nmid \Delta$, and E_p is an elliptic curve by Proposition 2.2. The reductions $\bar{T}_i = (\bar{e}_i, 0)$ are affine, so they differ from $\bar{O} = [0 : 1 : 0]$, and they are distinct, because $p \nmid e_i - e_j$ for $i \neq j$. So reduction is injective on the four points of $E[2](\overline{\mathbb{Q}})$.

Remark. Both conditions defining S are needed. If $p \mid e_i - e_j$, then $\bar{T}_i = \bar{T}_j$, so $T_i - T_j$, the third point of order two, reduces to $\bar{O}$. At $p = 2$, injectivity fails for every model of E: an elliptic curve over a field of characteristic 2 has at most two 2-torsion points over an algebraic closure [Sil09, Corollary III.6.4(c)]. Problem 3.1 shows that L can indeed ramify at 2.

(3) Let $p \notin S$, $Q \in E(\overline{\mathbb{Q}})$ with $2Q \in E(\mathbb{Q})$, and $\sigma \in I_p$. Show that $\sigma(Q) - Q$ reduces to $\bar{O}$, and deduce that $\sigma(Q) = Q$. Conclude that $L/\mathbb{Q}$ is unramified at p.

Solution. By Proposition 2.3 and $\bar{\sigma} = 1$,

$$\overline{\sigma(Q) - Q} = \bar{\sigma}(\bar{Q}) - \bar{Q} = \bar{Q} - \bar{Q} = \bar{O}.$$

By Problem 1.6(1), $\sigma(Q) - Q \in E[2](\overline{\mathbb{Q}})$, so $\sigma(Q) = Q$ by (2). So σ fixes the coordinates of Q, hence $\mathbb{Q}(Q)$. As this holds for every such Q, the element σ fixes their compositum L. Hence $I_p \subseteq \mathrm{Gal}_L$.

Remark. In the language of Problem 1.6(4), the homomorphism $\kappa(P, \cdot)$ is trivial on I_p: the class $\delta(P)$ is unramified at p. For general m and K the same argument works, with the injectivity of reduction on $E[m]$ for $v(m) = 0$ taken from the formal group [Sil09, Proposition VII.3.1(b)]. Item (2) is where rational 2-torsion makes this elementary.

(4) Show that $D \subseteq \mathbb{Q}(S, 2)$. Deduce that $\#\operatorname{Gal}(L/\mathbb{Q}) \leqslant 2^{s+1}$ and $\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant 4^{s+1}$.

Solution. $\boldsymbol{\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant 4^{s+1}}$**.** Let d represent a class of D, and let $p \notin S$, so p is odd, as $2 \in S$. Then $\sqrt{d} \in L$, and I_p fixes L by (3), so $\mathbb{Q}(\sqrt{d})$ is unramified at p, and $v_p(d)$ is even by Proposition 2.5. So $D \subseteq \mathbb{Q}(S, 2)$, an $\mathbb{F}_2$-vector space of dimension $s + 1$ (Definition 2.6). If $d_1, \ldots, d_n$, with $n \leqslant s+1$, represent a basis of D, then $L = \mathbb{Q}(\sqrt{d_1}, \ldots, \sqrt{d_n})$, and $[L : \mathbb{Q}] \leqslant 2^n \leqslant 2^{s+1}$. So $\mathrm{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^t$ with $t \leqslant s + 1$, and by Problem 1.6(3),

$$\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant \#\operatorname{Hom}\big((\mathbb{Z}/2\mathbb{Z})^t, (\mathbb{Z}/2\mathbb{Z})^2\big) = 4^t \leqslant 4^{s+1}.$$

Remark. This is the weak Mordell–Weil theorem for these curves. With the Mordell–Weil theorem, $E(\mathbb{Q}) \cong \mathbb{Z}^r \times E(\mathbb{Q})_{\mathrm{tors}}$, and $E(\mathbb{Q})_{\mathrm{tors}}/2$ has the same order as $E(\mathbb{Q})_{\mathrm{tors}}[2] = E[2](\overline{\mathbb{Q}})$, namely 4. So $\#E(\mathbb{Q})/2E(\mathbb{Q}) = 2^{r+2}$, and the bound says $r \leqslant 2s$.

3. An example

This section runs Problem 2.7 on $y^2 = x^3 - x$, that is, $(e_1, e_2, e_3) = (0, 1, -1)$. The images of its points of order two under the explicit form of the Kummer map are [Sil09, Proposition X.1.4].

Problem 3.1 (The curve $y^2 = x^3 - x$)**.**
In this problem, E is the curve $y^2 = x^3 - x$, with $T_1 = (0, 0)$, $T_2 = (1, 0)$ and $T_3 = (-1, 0)$. `Magma` finds its rank and its torsion subgroup, and checks that the halves of each T_i are defined over $\mathbb{Q}(\zeta_8)$:

```
E := EllipticCurve([-1, 0]);
Rank(E);                          // 0 true
TorsionSubgroup(E);               // Z/2 + Z/2
K<z> := CyclotomicField(8);
EK := ChangeRing(E, K);
[#DivisionPoints(EK ! [e, 0], 2) : e in [0, 1, -1]];   // [ 4, 4, 4 ]
```

So $E(\mathbb{Q}) = E[2](\overline{\mathbb{Q}})$.

(1) Compute S and the bound of Problem 2.7(4).

Solution. $\boldsymbol{S = \{2\}}$**, so** $\boldsymbol{\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant 16}$**.** The differences $e_i - e_j$ are ± 1 and ± 2, so $S = \{2\}$ and $s = 1$. Here $\mathbb{Q}(S, 2) = \langle -1, 2\rangle$, so $L \subseteq \mathbb{Q}(i, \sqrt{2})$, and the bound is $4^2 = 16$.

(2) For $i = 1, 2, 3$, find the four $Q \in E(\overline{\mathbb{Q}})$ with $2Q = T_i$, and show that $L = \mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$.

Solution. **The halves of** $\boldsymbol{T_1, T_2, T_3}$ **generate** $\boldsymbol{\mathbb{Q}(i)}$**,** $\boldsymbol{\mathbb{Q}(\sqrt{2})}$ **and** $\boldsymbol{\mathbb{Q}(\zeta_8)}$**.** A half $Q = (x, y)$ of a point of order two has order 4, so $y \neq 0$, and $y^2 = x^3 - x$. For T_1, $x(2Q) = 0$ gives $x^2 = -1$. Then $y^2 = x(x^2 - 1) = -2x$, and

$$Q = (i, \pm(1 - i)), \quad (-i, \pm(1 + i)),$$

as $(1-i)^2 = -2i$ and $(1+i)^2 = 2i$. For T_2, $x(2Q) = 1$ gives $(x^2+1)^2 = 4(x^3 - x)$, that is, $(x^2 - 2x - 1)^2 = 0$, so $x = 1 \pm \sqrt{2}$. Then $x^2 - 1 = 2x$, so $y^2 = 2x^2$ and $Q = (x, \pm\sqrt{2}\,x)$. For T_3, $x(2Q) = -1$ gives $(x^2 + 2x - 1)^2 = 0$, so $x = -1 \pm \sqrt{2}$. Then $x^2 - 1 = -2x$, so $y^2 = -2x^2$ and $Q = (x, \pm\sqrt{-2}\,x)$. As $E(\mathbb{Q}) = \{O, T_1, T_2, T_3\}$, the field L is the compositum of $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{-2})$, which is $\mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$, as $\zeta_8 = (1+i)/\sqrt{2}$. It is ramified at 2 only, as it should be.

(3) Let $\sigma, \tau \in \mathrm{Gal}(L/\mathbb{Q})$ with $\sigma(i) = -i$, $\sigma(\sqrt{2}) = \sqrt{2}$, $\tau(i) = i$ and $\tau(\sqrt{2}) = -\sqrt{2}$. Compute $\kappa(T_i, \sigma)$ and $\kappa(T_i, \tau)$ for $i = 1, 2, 3$, and deduce that no T_i lies in $2E(\mathbb{Q})$. Compute $\#E(\mathbb{Q})/2E(\mathbb{Q})$ and compare it with the bound of (1).

Solution. **$\#E(\mathbb{Q})/2E(\mathbb{Q}) = 4$, against the bound 16.** By bilinearity it is enough to treat T_1 and T_2, as $T_3 = T_1 + T_2$. Take the half $Q = (i, 1-i)$ of T_1. It is fixed by τ, so $\kappa(T_1, \tau) = O$. And $\sigma(Q) = (-i, 1+i)$, while $-Q = (i, i-1)$. The line through these two points has slope $(i - 1 - 1 - i)/(2i) = i$, so $\sigma(Q) - Q$ has x-coordinate $i^2 - (-i) - i = -1$, and $\kappa(T_1, \sigma) = T_3$. Take the half $Q = (1+\sqrt{2}, 2+\sqrt{2})$ of T_2. It is fixed by σ, so $\kappa(T_2, \sigma) = O$. And $\tau(Q) = (1-\sqrt{2}, 2-\sqrt{2})$, while $-Q = (1+\sqrt{2}, -2-\sqrt{2})$. The slope is $-4/(2\sqrt{2}) = -\sqrt{2}$, so $\tau(Q) - Q$ has x-coordinate $2 - (1-\sqrt{2}) - (1+\sqrt{2}) = 0$, and $\kappa(T_2, \tau) = T_1$. In all,

	σ	τ
T_1	T_3	O
T_2	O	T_1
T_3	T_3	T_1

No row vanishes, so no T_i lies in $2E(\mathbb{Q})$, by Problem 1.6(3). As $E(\mathbb{Q}) = E[2](\overline{\mathbb{Q}})$, also $2E(\mathbb{Q}) = \{O\}$ directly. Hence $\#E(\mathbb{Q})/2E(\mathbb{Q}) = 4$. The pairing is perfect, as it should be: $\mathrm{Gal}(L/\mathbb{Q})$ has order 4, and only 4 of the 16 homomorphisms $\mathrm{Gal}(L/\mathbb{Q}) \to E[2](\overline{\mathbb{Q}})$ come from rational points.

Remark. The explicit Kummer map of [Sil09, Proposition X.1.4] sends $P = (x, y) \notin \{T_1, T_2\}$ to $(x - e_1, x - e_2) \in \mathbb{Q}(S, 2)^2$, with a modified formula at T_1 and T_2. Here it gives $T_1 \mapsto (-1, -1)$, $T_2 \mapsto (1, 2)$ and $T_3 \mapsto (-1, -2)$. Each coordinate records the action of $\mathrm{Gal}(L/\mathbb{Q})$ on a square root, which matches the table: $\kappa(T_1, \cdot)$ sees only $\sqrt{-1}$, $\kappa(T_2, \cdot)$ only $\sqrt{2}$. Deciding which of the 16 classes of $\mathbb{Q}(S, 2)^2$ come from rational points is the descent problem of [Sil09, §X.1].

References

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